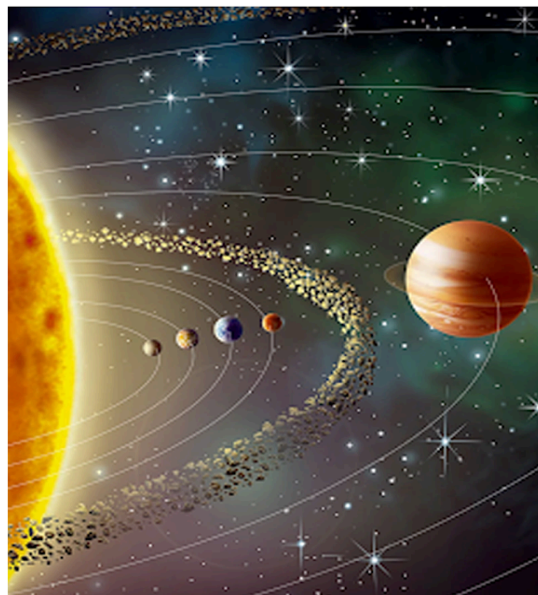
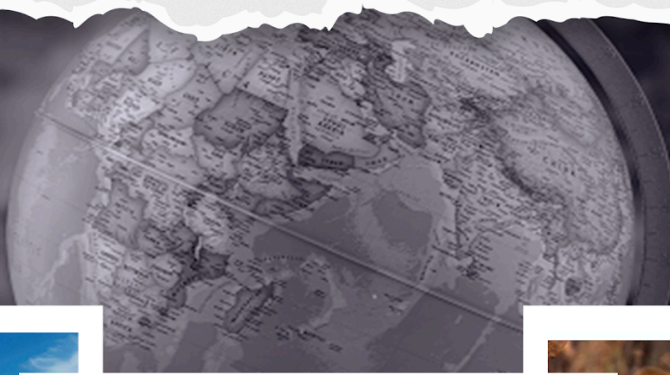


# **General Knowledge**

## **World Geography**

### **STUDY NOTES**

**Climatology: Solar Radiation, Heat  
Balance and Temperature**



# Climatology: Solar Radiation, Heat Balance and Temperature

Climatology is the study of atmospheric conditions over a longer period of time. It includes the study of different kinds of weather that occur at a place. Dynamic change in the atmosphere brings about variation and occasionally great extremes that must be treated on the long term as well as the short term basis.

## Solar Radiation

### Insolation:

- The solar energy radiated from the outer surface of the sun is called **electromagnetic radiation**.
- The solar energy received at the earth's surface is called **insolation or solar radiation**.
- **The globe is divided into 3 zones** on the basis of the amount of insolation received during the course of a year:
  1. **Low Latitude/ Tropical Zone:**
    - extends **between the tropics of Cancer and Capricorn** where the sun's rays are vertical twice a year.
    - receives the **highest amount of insolation** and exhibits little seasonal variation.
  2. **Middle Latitude Zone:**
    - extends **between 23.4° and 66° latitudes** in both hemispheres where every place receives maximum and minimum insolation once a year.
    - Insolation is never absent at any time of the year but **seasonal variation** increases with increasing latitudes.
  3. **Polar Zone:**
    - extends **between 60° and 90° (poles) latitudes** in both hemispheres where every place receives maximum and minimum insolation during a year.
    - **Insolation becomes zero** due to the absence of direct sun rays.

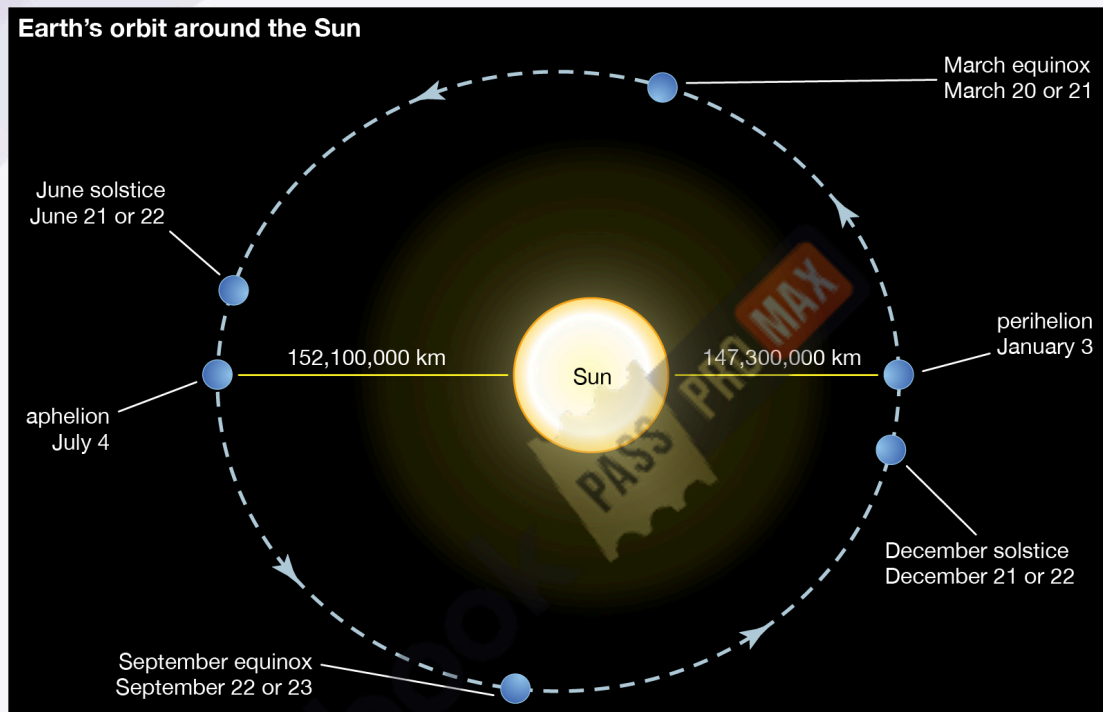
### Variability of Insolation at the surface of the Earth

The insolation received on earth is not the same everywhere. The amount of insolation varies from the equator to the poles due to certain astronomical and geographical factors:

### 1. Revolution of Earth around the sun:

The distance between the sun and the earth changes during the course of a year because the earth revolves around the sun in an **elliptical orbit**.

#### Aphelion and Perihelion:

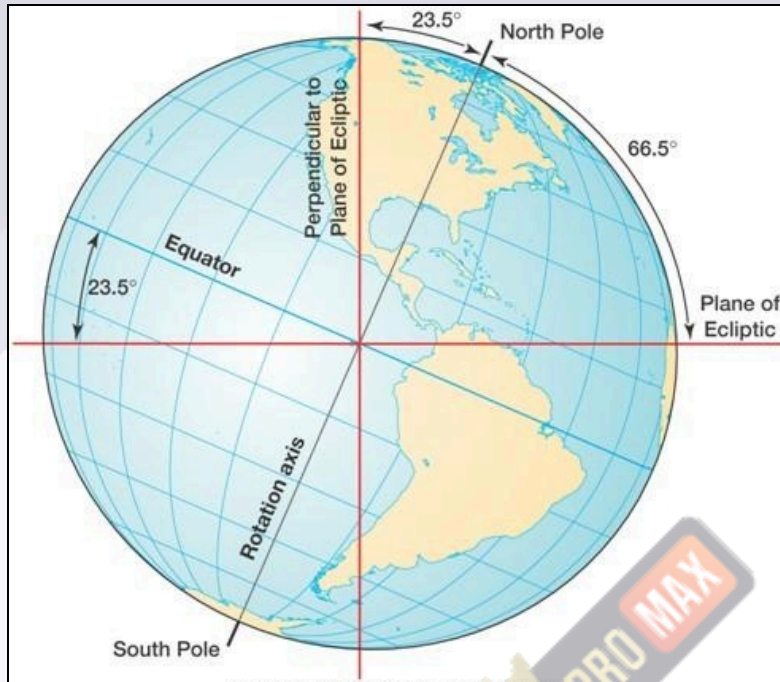


- At the time of **perihelion** on January 3, the **earth is nearest to the sun**.
- At the time of **aphelion** on July 4, the earth is **farthest away from the sun**.
- Therefore, the **annual insolation received by the earth at perihelion is slightly more than the amount received at aphelion**.
- However, the effect of this variation in the solar output is marked by other factors **like the distribution of land and sea and the atmospheric circulation**. Hence, this variation in the insolation does not have a great effect on daily weather changes on the surface of the earth.

### 2. Rotation of earth on its axis:

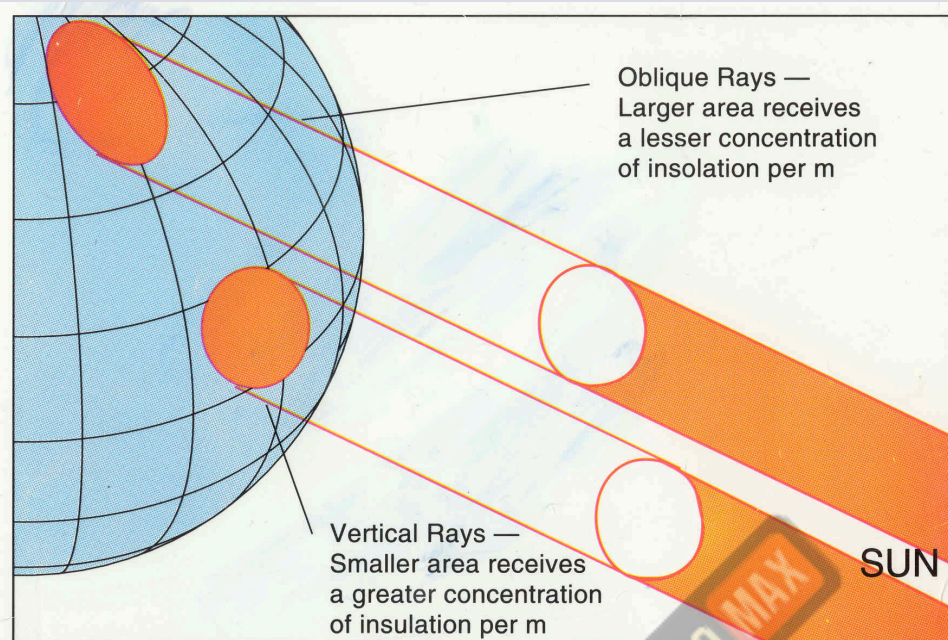
- The earth rotates around its axis and makes an **angle of  $66\frac{1}{2}^\circ$**  with the plane of its orbit around the sun affecting the amount of insolation received at different latitudes.
- The seasons in each hemisphere are dictated not by the closeness to the sun but by the **axial tilt of the earth**.





### 3. The angle of inclination of the sun's rays:

- The angle between the rays of the sun and the tangent to the surface of the earth at a given place determines the insolation received at that place.
- The angle formed by the sun's ray with the tangent of the earth's circle at a point is called the **angle of incidence**.
- The sun's rays are more or less vertical at the equator and become oblique towards the poles. [**Vertical rays = more insolation**]

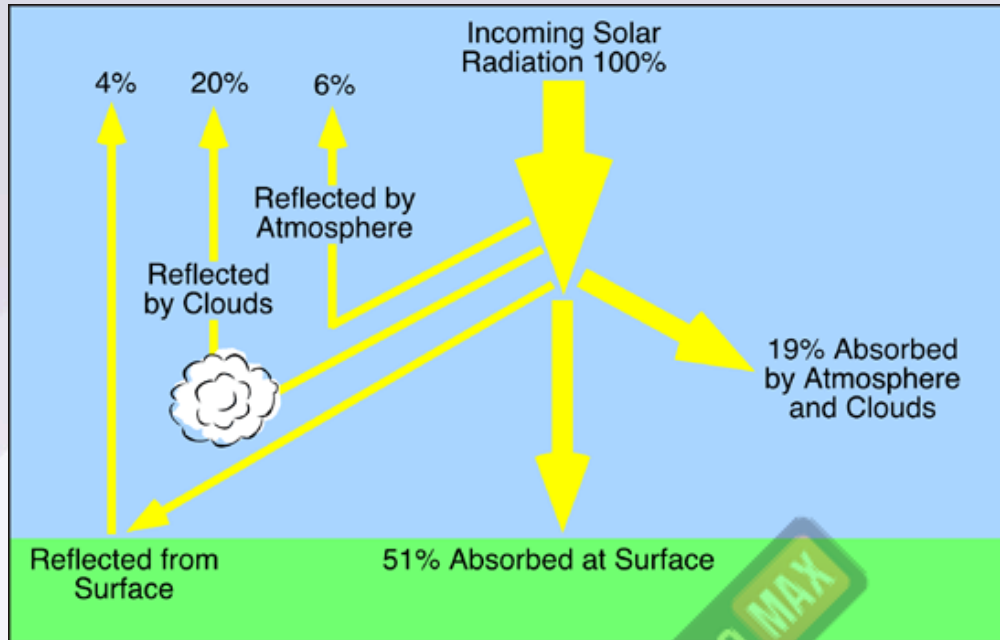


#### 4. Length of day:

- The duration of day is controlled partly by latitude and partly by the season of the year.
- The longer duration of sunshine (length of day) and shorter duration of night enable the ground to receive a larger amount of insolation.
- Despite the longest length of day at the poles, insolation **becomes minimal** because: **the sun's rays become more or less parallel to the ground surface and the ice cover reflects most of the solar radiation.**
- Thus it is apparent that the **angle of the sun's rays controls the insolation more effectively than the length of day.**

#### 5. Transparency of the atmosphere:

- The earth's atmosphere is more or less **transparent to shortwave solar radiation** which has to pass through the atmosphere before striking the earth's surface.
- The incoming short-wave radiation undergoes the following processes in the atmosphere:



- **Absorbed** - 19% is absorbed selectively by the atmospheric gases (ozone, oxygen, and carbon dioxide), water vapor, haze, etc and 51% by the earth's surface.
- **Scattered** - 23% is scattered by haze and dust particles.
- **Reflected** - 20% is reflected back to space by high clouds and 4% by ground surface.

**Albedo:** The portion of incident radiation energy reflected back from a surface is called albedo.

#### 6. Sunspots:

- Sunspots are created on the **outer surface of the sun** due to periodic disturbances and explosions.
- The number of sunspots varies in a **cycle completed in 11 years**.
- The energy radiated from the sun increases when the number of sunspots increases and therefore the amount of insolation received also increases.

#### 7. Topographical variations:

- Topographical variations are the major factors modifying the distribution of insolation.
- Variability in **elevation, surface orientation** (slope and aspect), and obstruction by surrounding topographic features create strong local gradients of insolation.
- **Maximum insolation is received over the subtropical deserts.** The equator receives comparatively less insolation than the tropics due to the presence of clouds.

- Generally, at the same latitude, the insolation is more over the continent than over the oceans **because more clouds over the oceans reflect sun rays back into space.**

**Isohels:** These are lines connecting points on the earth's surface that receive **equal amounts of sunshine.**

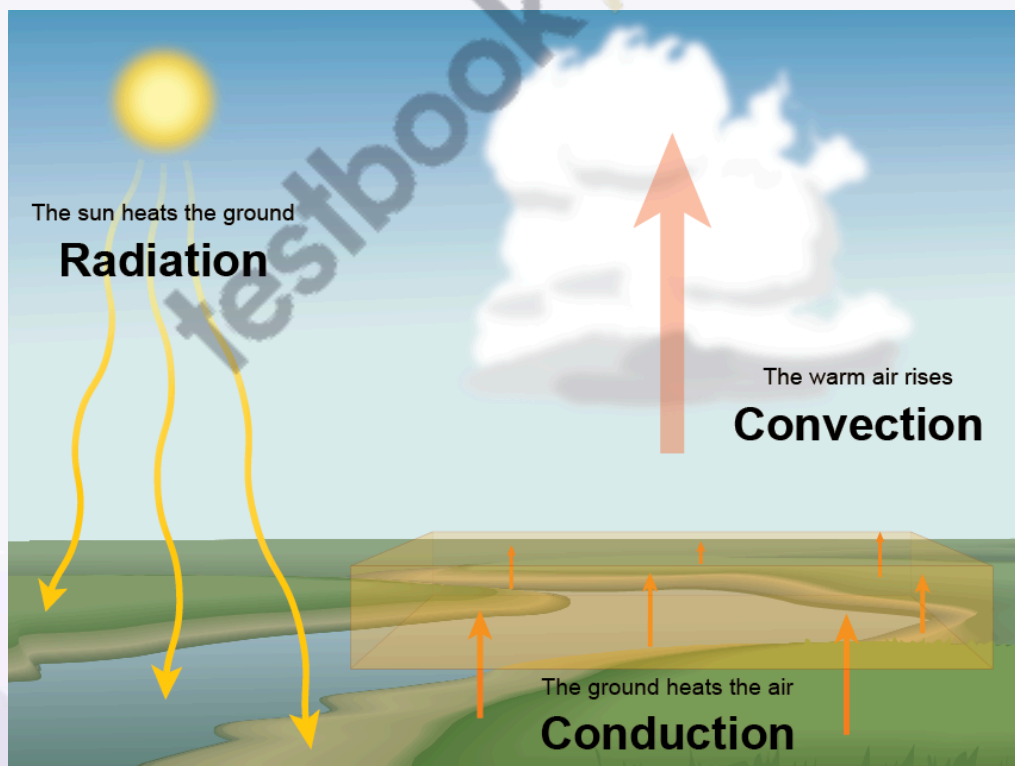
## Heat Balance

### Heating and Cooling of the Atmosphere

The Sun is the ultimate source of atmospheric heat and energy.

- The process of radiation of heat energy from the earth's surface is called **ground radiation** (including radiation from both ground and water surface.).
- The process of radiation of terrestrial heat energy from the atmosphere back to the earth's surface is called **counter-radiation.**

**The common types of energy transport involved in heating of the atmosphere are:**





### i. Conduction:

- The transfer of heat **through the molecules of matter** in anybody is called conduction.
- Accomplished by the transfer of heat from one part of the body to the other part of the same body and from one body to the other touching body.
- The flow continues till the temperature of both the objects becomes equal or the contact is broken.
- The conduction in the atmosphere occurs at the **zone of contact between the atmosphere and the earth's surface**. However, this is a minor method of heat transfer in terms of warming the atmosphere since it only affects the air close to the earth's surface. This is because of the fact that the air is a poor conductor of heat.

### ii. Convection:

- The transfer of heat energy through the **movement of molecules** from one place to another **within the same body** is called convection.
- The air in contact with the earth rises vertically on heating in the form of currents.
- The heating of the air leads to its expansion. Its density decreases and it moves upwards.
- Continuous ascent of heated air creates a vacuum in the lower layers of the atmosphere. As a consequence, cooler air comes down to fill the vacuum. This process of **vertical heating of the atmosphere** is known as convection.
- The convective transfer of energy is **confined only to the troposphere**.

### iii. Advection:

- The transfer of heat through **horizontal movement of air** is called advection. These winds take the **characteristics of their source of origin** with them. The temperature of a place will rise if it lies on the path of winds coming from warmer regions and the temperature will fall if the place lies on the path of the winds blowing from cold regions.
- The horizontal movement of the air is relatively **more important than the vertical movement**.
- e.g., 'Loo' of north India is a hot wind, and 'Sirocco' is also a hot wind that carries heat from the Sahara desert to the Mediterranean regions.
- In middle latitudes, most diurnal (day and night) variation in daily weather is caused by advection alone.

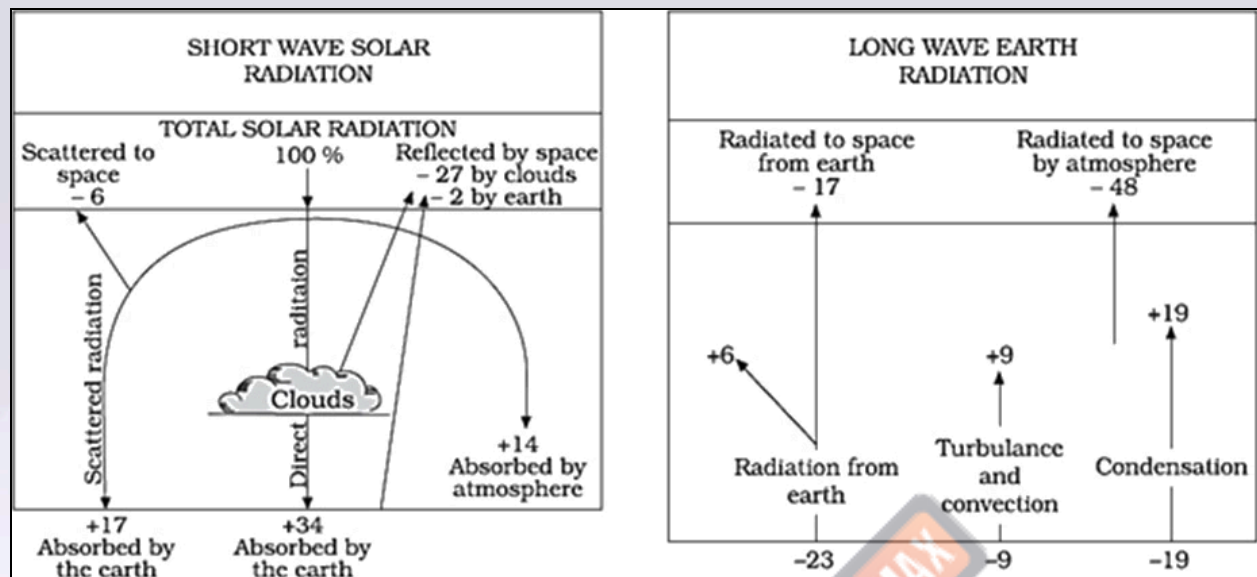


### Terrestrial Radiation:

- The **insolation received by the earth is in short waveforms** and heats up its surface.
- The earth after being heated becomes a radiating body and it radiates energy to the atmosphere in the long waveform.
- This energy **heats up the atmosphere from below**. This process is known as terrestrial radiation.
- The long wave radiation is absorbed by the atmospheric gases, particularly by carbon dioxide and the other greenhouse gases. Thus, the **atmosphere is indirectly heated by the earth's radiation**.
- The atmosphere in turn radiates and transmits heat to space. Finally, the amount of heat received from the sun is returned to space, thereby maintaining constant temperature at the earth's surface and in the atmosphere.

### Heat Budget of the Earth

- The earth as a whole **does not accumulate or lose heat**. It maintains its temperature.
- This can happen only if the amount of heat received in the form of insolation equals the amount lost by the earth through terrestrial radiation.
- Consider that the insolation received at the top of the atmosphere is 100 percent.
- While passing through the atmosphere some amount of energy is reflected, scattered, and absorbed. Only the remaining part reaches the earth's surface.
- **Roughly 35 units are reflected back to space even before reaching the earth's surface.** Of these, 27 units are reflected back from the top of the clouds and 2 units from the snow and ice-covered areas of the earth. The **reflected amount of radiation is called the albedo** of the earth.
- **The remaining 65 units are absorbed**, 14 units within the atmosphere and 51 units by the earth's surface.
- **The earth radiates back 51 units in the form of terrestrial radiation.** Of these, 17 units are radiated to space directly and the remaining 34 units are absorbed by the atmosphere (6 units absorbed directly by the atmosphere, 9 units through convection and turbulence, and 19 units through latent heat of condensation).



- **48 units absorbed by the atmosphere** (14 units from insolation +34 units from terrestrial radiation) are also radiated back into space.
- Thus, the total radiation returning from the earth and the atmosphere respectively is **17+48=65** units which balances the total of 65 units received from the sun.
- This is termed the **heat budget or heat balance** of the earth.
- This explains why the earth neither warms up nor cools down despite the huge transfer of heat.

## Temperature:

- The interaction of insolation with the atmosphere and the earth's surface creates heat which is measured in terms of temperature.
- While heat represents the molecular movement of particles comprising a substance, temperature is the measurement in degrees of **how hot (or cold) a thing (or a place) is**.

## Factors controlling temperature distribution:

The temperature of air at any place is influenced by the following factors:

### The latitude:

- The temperature of a place depends on the insolation received.

- The **insolation varies according to the latitude** hence the temperature also varies accordingly.

#### **The altitude:**

- The atmosphere is indirectly heated by terrestrial radiation from below. Therefore, the places near the sea level record higher temperatures than the places situated at higher elevations.
- **Temperature generally decreases with increasing height.**
- The rate of decrease of temperature with height is termed as the **normal lapse rate**. It is  $6.5^{\circ}\text{C}$  per 1,000 m.

#### **Distance from the sea:**

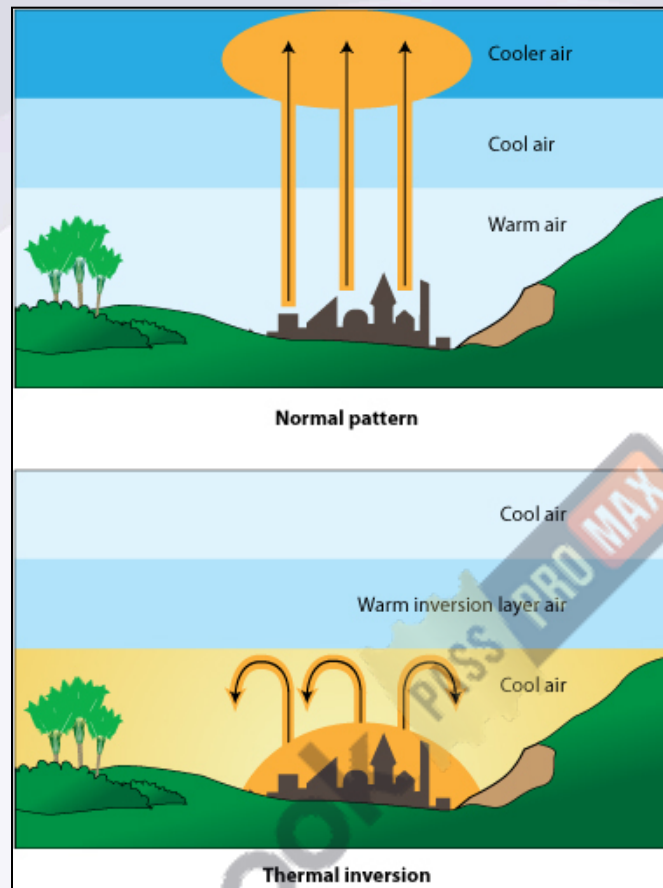
- The location of a place with respect to the sea also determines the temperature of that place.
- Compared to land, the sea gets heated slowly and loses heat slowly. Land heats up and cools down quickly. Therefore, the **variation in temperature over the sea is less compared to land.**
- The places situated near the sea come under the moderating influence of the sea and land breezes which moderate the temperature.

#### **Air-mass and Ocean currents:**

- The places, which come under the influence of **warm air masses experience higher temperatures** and the places that come under the influence of **cold air masses experience low temperatures.**
- Similarly, the places located on the coast where the **warm ocean currents flow record higher temperatures** than the places located on the coast where the cold currents flow.

**Isotherms:** These are lines joining places having **equal temperatures**.

## Inversion of Temperature:



- Normally, temperature decreases with an increase in elevation. It is called the **normal lapse rate**.
- At times, the situation is reversed and the **normal lapse rate is inverted**. It is called the Inversion of temperature.
- Inversion is usually of **short duration** but quite common nonetheless.
- A long winter night with **clear skies and still air** is an ideal situation for inversion.
- The heat of the day is radiated off during the night and by early morning hours, the earth is cooler than the air above.
- **Over polar areas, temperature inversion is normal throughout the year.**
- **Surface inversion** promotes stability in the lower layers of the atmosphere. Smoke and dust particles get collected beneath the inversion layer and spread horizontally to fill the lower strata of the atmosphere.
- **Dense fog** in the mornings are common occurrence, especially during winter. This inversion commonly lasts for a few hours until the sun comes up and begins to warm the earth.



- **The inversion takes place in hills and mountains due to air drainage.**
- Cold air in the hills and mountains, produced during the night, flows under the influence of gravity.
- Being heavy and dense, the cold air acts almost like water and moves down the slope to pile up deeply in pockets and valley bottoms with warm air above. This is called **air drainage**.
- **It protects plants from frost damage.**